Inha University in Tashkent
School of Computer Science and Information Engineering
Course Name: Object Oriented Programming 2
Semester and Year: Spring Freshman, 2026
Laboratory Assignment 4: Object-Oriented Programming in C++

Note: 1. Submission only via E-CLASS is acceptable (If you have any issue with E-CLASS, Please consult IT Dept.)

2. Comments should be included in your source code.

3. Do not plagiarize or use AI tools to complete the assignment. If a student is caught doing so, he/she will receive ZERO for the assignment.

4. Upload your completed laboratory exercise as a zip file including the appropriate files **(.cpp, and .h),** to the eClass system before the deadline. The zip file must be named with your **ID number.**

Exercise 4: Stock exchange system

Objective: Develop a system that simulates a stock exchange by managing buying, selling, and limit orders. The system should display live stock prices and market trends using operator overloading.

Tasks:

1. Define a class **StockEx**.
2. Declare the following private members:

- currency (string) – Represents the currency name.
- mktrate (float) – Represents the market rate of the currency.
- date (string) – Represents the date of the market rate.

3. Implement a parameterized constructor to initialize stock exchange details (currency, mktrate, date).

4. Overload the multiplication operator (*) to exchange currency based on the market rate.
5. Overload the subtraction operator (-) to calculate the difference between two market rates.
6. Overload the stream insertion operator (<<) to print stock exchange details.
7. Overload the stream extraction operator (>>) to take user input for a StockEx object.
8. Overload the relational operator (>) to compare two MarketRate objects.
9. Create StockEx objects, use overloaded operators and display the results.

Output:

```
Enter details for first market:
Enter currency: USD
Enter market rate: 12100
Enter date: 12.03.2026

Enter details for second market:
Enter currency: EUR
Enter market rate: 13200
Enter date: 12.03.2026

First Market:
Currency: USD
Market Rate: 12100
Date: 12.03.2026

Second Market:
Currency: EUR
Market Rate: 13200
Date: 12.03.2026

Choose market for currency exchange:
1. First Market
2. Second Market
Enter your choice: 2
Enter amount to exchange: 10
Converted amount: 132000

Difference between market rates: -1100
Second market rate is higher .
```